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EU AI Act

Luxembourg

AI Ecosystem Brief

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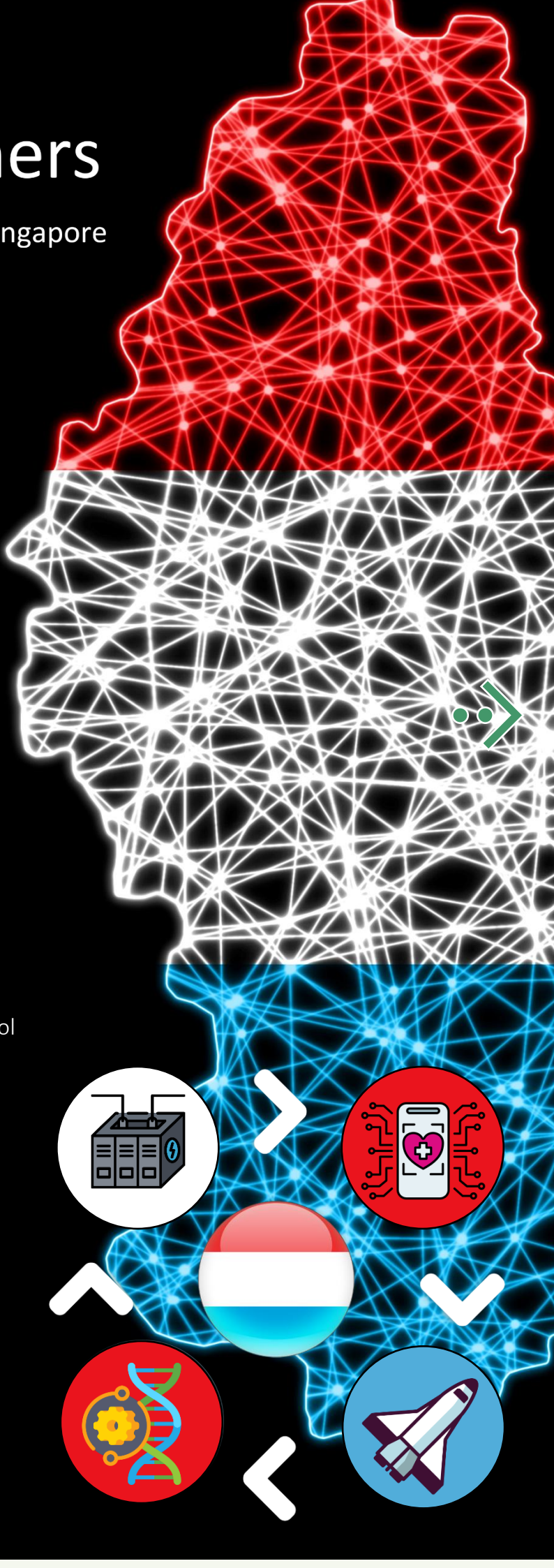
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Who Are We

AI That You Can Trust

Why Us?

Stay on the right side of history. At AI & Partners, we believe AI should unlock potential—not cause harm. We've seen the fear and fallout when teams lose control of AI, but also the trust and innovation that follow when it's handled responsibly. That's why we exist: to help you build AI you can trust and stand behind—for the long run.

80%

of AI systems
are unknown

What Do We Do?

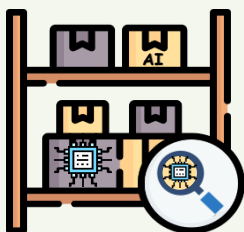
We enable safe AI usage—for your organization and your clients. Unknown AI adoption leads to confusion, risk, and reputational damage. We help you take control with tools to identify, monitor, and govern all AI systems—so you're not reacting to AI, you're leading it.



How Do We Do It?

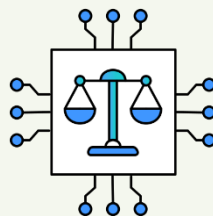
Do you know what AI systems you have? Identify all known and unknown AI systems (algorithms, LLMs, prompts, and models) from all internal and external AI vendors, automated by generating your inventory. Overall, 80% of AI inventory is unknown to our clients.

How do you guarantee ongoing safe AI use? Continuously monitor deployed AI systems for performance drift, anomalies or failures, real-world impacts, and emerging risks (e.g. data poisoning). Any malfunction of an AI system has severe implications for organisations (e.g. inability to assess online misinformation that leads to widespread public mistrust), so monitoring becomes a matter of urgency.



AI Discovery & AI Inventory

Automatically detect all AI systems, including models, algorithms, and prompts, and maintain a live, always-updated register for full visibility and compliance.



Responsible AI

Embed fairness, transparency, and control into every stage of AI use—aligning with the EU AI Act and building 'Trustworthy-by-Design'.



Model Monitoring

Continuously track your AI models after deployment to detect drift, bias, or failure—so you stay in control and prevent harm before it happens.



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Overview

Artificial Intelligence (AI) is reshaping economies and governance worldwide, and Luxembourg is establishing itself as a nimble advocate of ethical, human-centric AI within the European Union. Anchored in the EU's evolving regulatory landscape, Luxembourg has adopted a supervisory-first, alignment-driven approach that emphasises rights-based safeguards, regulatory “living-lab” experimentation and cross-border cooperation. Rather than introducing standalone AI legislation, Luxembourg is leveraging its 2019 Artificial Intelligence: A Strategic Vision, the 2025 Digital-Sovereignty 2030 update and other soft-law instruments to promote trustworthy AI development.

At the heart of Luxembourg's evolving AI-governance strategy is Bill 8476, which designates the Commission nationale pour la protection des données (CNPD) as horizontal market-surveillance authority and single point of contact under the EU AI Act. Sectoral watchdogs—including the CSSF (finance), ILR (connectivity & media), CAA (insurance) and ALMPS (health)—round out the list of competent authorities. A distinctive feature of Luxembourg's approach is its “Sandkëscht” AI sandbox, launched in May 2024, which lets public- and private-sector teams pilot high-risk AI systems in a GDPR-aligned environment. These regulatory tools are complemented by strategic resources such as the Digital Decade National Strategic Roadmap, which together articulate a vision for safe AI.

Luxembourg's governance model is underpinned by a multi-actor institutional architecture. The CNPD leads on horizontal supervision, while sector-specific regulators oversee domain use. The AI4Gov committee coordinates public-sector pilots and capacity-building, and research centres like the University of Luxembourg's SnT, LIST and LIH channel more than €200 million in National Research Fund (FNR) grants into applied-AI projects. This whole-of-society approach embeds AI governance within national goals.

Luxembourg also shows strong commitment to international standards. The country aligns closely with EU initiatives, participates in EuroHPC, CLAIRE and the Hiroshima AI Process, and co-funds Franco-German-Luxembourg research corridors. Such engagement ensures that domestic practices remain interoperable and globally relevant, reinforcing Luxembourg's image as a constructive contributor to human-centric AI governance.

Aligned with its strategic roadmaps, Luxembourg is targeting finance, health-tech, space, smart mobility and clean-tech for AI-enabled transformation. Flagship initiatives include the LHoFT “AI Experience Centre” for sovereign financial-AI pilots, precision-medicine pipelines under the Health Directorate and the 215 km Schengen cross-border autonomous-driving corridor overseen by ILR. Education and talent programmes—such as the Digital Skills Bridge, Luxembourg Tech School and AI-focused fellowships—help cultivate a skilled, future-ready workforce.

Luxembourg's data and infrastructure capabilities also reinforce its AI readiness. Investments in the MeluXina EuroHPC supercomputer (to be upgraded with an AI-optimised cluster by 2026), sovereign GovCloud services and the Luxembourg National Data Service align with EU instruments like the Data Governance Act and the Data Act. Sectoral data spaces for health, energy and finance, coupled with near-universal 5 G/VHCN coverage, underscore a future-facing commitment to scalable, trustworthy data use. Through this combination of EU-aligned oversight, living-lab experimentation and globally harmonised ethics frameworks, Luxembourg is building a comprehensive yet agile model of AI governance.





Specific AI Governance or Law

Luxembourg is implementing the EU AI Act through Bill 8476, designating the CNPD as lead authority and coordinator. Sector-specific oversight is assigned to bodies like CSSF and CAA. A national AI sandbox, "Sandkëscht," launched in May 2024 to support GDPR-aligned AI testing ahead of 2026 obligations.

Specific AI Governance in Luxembourg

The AI Act entered into force on 1 August 2024, requiring each Member State to publish – by 2 November 2024 – a list of competent authorities. Luxembourg responded with Bill 8476 (23 December 2024), which will: Designate the Commission nationale pour la protection des données (CNPD) as the default market-surveillance authority and sole notified body for high-risk AI used by law-enforcement, immigration and asylum agencies; Appoint OLAS and ALMPS as notifying authorities for general-purpose and medical AI, respectively; Allocate sectoral oversight to the CSSF (finance), CAA (insurance), ILR (essential-service operators), the Judicial Supervisory Authority and others, all coordinated by the CNPD. The listed bodies will gain new powers—from documentation requests to coordinated inspections—once the Act's high-risk provisions start applying on 2 August 2026. Regulatory sandboxes. To prepare industry for compliance, the CNPD launched the "Sandkëscht" AI sandbox on 21 May 2024. The programme lets public- and private-sector teams test AI systems in a controlled, GDPR-aligned environment and receive regulator feedback before market entry.

AI Ethics Guidelines and Sectoral Governance Tools

Luxembourg applies a layered model: the EU AI Act on top, reinforced by GDPR, cybersecurity, consumer-protection and sector rules. Key national instruments include:

- Human-centric Vision (2019): Puts individual rights "in control of AI" and prioritises seven focus areas—data stewardship, ethics & privacy, lifelong learning, public-sector AI, etc.
- Accelerating Digital Sovereignty 2030 (2025): Updates targets and budgets through 2030; anchors AI within a tri-strategy of Data–AI–Quantum.

Proposed AI Regulatory Authority and Future Outlook

With Bill 8476 close to parliamentary adoption, the CNPD-centred oversight constellation, "Sandkëscht" sandboxing and massive MeluXina-AI investment form an adaptive, EU-compliant governance stack. Results from sandbox pilots and flagship projects will feed directly into secondary legislation and technical guidelines, ensuring Luxembourg's framework evolves in lock-step with the AI Act and local socio-economic priorities. In intertwining human-rights safeguards, data-sovereign infrastructure and real-world experimentation, Luxembourg is carving out a compact yet influential model for trustworthy, innovation-driven AI in Europe.



Discussion: Luxembourg's National AI Strategy

Luxembourg's AI-governance approach is grounded in a value-driven, EU-aligned framework that merges ethical safeguards, public-sector leadership and strategic infrastructure investment to accelerate responsible AI adoption. The April 2024 Digital Decade National Strategic Roadmap for Luxembourg (hereafter "the Roadmap") operationalises this vision by mapping concrete trajectories and policy levers to meet — and, in some areas, exceed — the EU's 2030 Digital Decade targets.

To be among the most advanced digital societies in the world, especially in the EU

To become a data-driven and sustainable economy

To support human-centric AI development

Analysis

Luxembourg's AI Roadmap is grounded in a human-centric, rights-based approach that echoes the country's 2019 AI Strategic Vision. It reaffirms that the digital transition must prioritize individuals, embedding fairness, transparency, and privacy into every policy and measure. This philosophy aligns closely with EU and OECD guidelines and supports Luxembourg's decision to apply the EU AI Act directly, rather than introducing separate national legislation. The Roadmap sets evidence-led policy targets, diagnosing both strengths and areas for improvement. On digital skills, 64% of residents already possess at least basic competencies, but the country must still close a 16-percentage-point gap to reach the EU's 80% target by 2030. Luxembourg's infrastructure is strong, with 93% coverage in both 5G and very high-capacity networks (VHCN), positioning it as a frontrunner within the EU. In business adoption, Luxembourg's SMEs outperform the EU average in AI usage (13% vs. 8%) and big data analytics (19% vs. 14%), yet overall advanced technology adoption must triple to meet the 75% target. These figures inform a governance model calibrated to risk and readiness: high connectivity enables ambitious AI pilots, while lower SME digital intensity calls for targeted support.

To ensure inclusive, context-sensitive governance, Luxembourg has implemented a National Action Plan for Digital Inclusion, featuring 40 initiatives alongside custom adult learning programs to help disadvantaged groups access digital services and participate in online civic life. Responsibility for digital governance is broadly distributed among developers, government ministries, and civil society, with central coordination maintained by the Ministry for Digitalisation and the Interministerial Committee on Digital Policies. While the Roadmap does not explicitly use the term "AI Sandboxes," it mandates that each competent authority provide controlled testing environments for high-risk AI and edge-cloud technologies. These are supported by fiscal incentives such as 18% tax credits on both capital and operational expenditures related to digital transformation, particularly benefiting SMEs.

The Roadmap also outlines strategic instruments and institutional support across key pillars. For data and infrastructure sovereignty, institutions like the Luxembourg National Data Service (LNDS) and the MeluXina supercomputer facilitate secure, public-sector data sharing and AI training while contributing to EuroHPC's quantum capacity. Business transformation is aided by initiatives such as Fit 4 Digital vouchers, tax credits, and an expanded Digital Innovation Hub targeting the data economy. Public-sector leadership is demonstrated through projects like the MyGuichet.lu e-government platform and IT infrastructure upgrades that pilot sovereign large language models (LLMs) while upholding GDPR standards. On the skills front, the rebooted Digital Skills Platform and the National Action Plan for Digital Inclusion promote coding, AI literacy, and upskilling with attention to gender and cross-border worker disparities. In the area of standards development and global engagement, Luxembourg is taking active roles in ISO/IEC AI standardization through ILNAS, and is participating in international test beds such as 5G DeLux (a 98 km corridor with Germany) and 5G Melusina (a rail corridor with France). These projects support EU spectrum and safety standards through seamless handover demonstrations. Additionally, MeluXina's upcoming AI-optimized upgrade and quantum-simulator add-on will link Luxembourg's computing strategy with EuroHPC's wider supercomputing ecosystem. Looking ahead, the Roadmap's combination of EU-compliant regulation, data-driven targets, and sandbox-enabled experimentation positions Luxembourg to harness AI's economic and societal benefits while maintaining public trust. Should key measures scale successfully—such as increasing SME AI use from 13% to 75% and raising digital skills to 80%—Luxembourg is set to meet its Digital Decade goals and solidify its role as a compact yet influential "living laboratory" for trustworthy AI in Europe.





Luxembourg's Approach to AI Regulation and Governance

Luxembourg has positioned itself as a nimble, living-lab partner in shaping AI governance inside the EU, coupling rigorous rights-based safeguards with fast-cycle experimentation. Without creating a standalone AI statute, the Grand Duchy advances responsible, human-centric AI through strategic visions, regulatory sandboxes and sovereign-data infrastructure that together foster a resilient, innovation-ready ecosystem.

1 Regulatory and Strategic Frameworks

Direct application of the EU AI Act. A draft law approved on 29 November 2024 (Bill 8476) designates the CNPD as horizontal market-surveillance authority and single point of contact, while naming sector bodies such as the CSSF, ILR, ALMPS and CAA for domain-specific oversight. The bill obliges every authority to offer an AI regulatory sandbox and equips them with inspection and sanction powers that kick in on 2 August 2026.

Luxembourg's 2019 Artificial Intelligence: A Strategic Vision sets three ambitions—advanced digital society, data-driven economy, human-centric AI—and commits to regular updates, most recently through the broader Digital-Sovereignty 2030 programme.

Living-lab sandboxes. The vision paper explicitly tasks government with “studying the feasibility of regulatory sandboxes,” an idea now enshrined in Bill 8476 and operationalised by the CNPD's “Sandkëscht” sandbox launched in May 2024.



2 Institutional Support and Strategic Missions

- **Multi-actor oversight.** Alongside the CNPD, Luxembourg relies on sectoral watchdogs—CSSF (finance), ILR (connectivity & media), ALMPS (health), CAA (insurance)—co-ordinated through the CNPD's forthcoming National AI Coordination Hub.
- **Public-sector task force.** The AI4Gov committee unites ministries and agencies to scout, pilot and scale AI projects across government and to build in-house expertise on procurement and ethics.
- **Research & funding engines.** The FNR has channelled ~€200 million into AI-related research over five years, while funds such as the Digital Tech Fund and SNCI provide venture finance for AI start-ups.

3 Standards Development and Global Engagement

Luxembourg embeds EU guidance—e.g., the Ethics Guidelines for Trustworthy AI—and participates in ISO/IEC norm-setting via ILNAS. The strategy calls for intensified Franco-German-Luxembourg collaboration through a cross-border research cluster and active roles in initiatives like EuroHPC and the Hiroshima AI Process, keeping national practice interoperable with global norms.

4 Sectoral and Extended Regulatory Approaches

- **Data protection.** The CNPD enforces GDPR and, under Bill 8476, gains new AI-specific powers, including documentation requests and coordinated inspections.



- **Finance & insurance.** The CSSF is rolling out sovereign cloud pilots (Clarence) for confidential AI analytics, while the draft law extends CSSF/CAA mandates to high-risk AI in finance and insurance.
- **Connectivity & mobility.** ILR oversees AI use in networks and runs the 215 km Schengen cross-border autonomous-driving corridor with France and Germany as a real-world test site.

5 Fostering Innovation and Indigenous Development

The living-laboratory model prioritises applied research, with Luxinnovation's Digital Innovation Hub and LHoFT's "AI Experience Centre" funnelling SMEs into sandbox trials and grant schemes. Funding tools like the Future Fund de-risk early AI ventures, while FNR programmes spur spin-offs such as LuxAI and DataThings.

6 AI in Public-Sector Transformation

Government modernisation hinges on AI-enabled Digital by Default services, expansion of the national open-data portal and agency pilots ranging from multilingual chatbots on Guichet.lu to sovereign LLMs for legal drafting. The AI4Gov roadmap scores projects on feasibility and public value to ensure ethical, citizen-centric deployment.

7 AI Skills, Education and Inclusion

Measures include the Digital Skills Bridge upskilling scheme, AI modules across secondary-school curricula, and targeted fellowships to attract female researchers. The strategy pledges lifelong-learning programmes so that workers and the unemployed can transition into AI-augmented roles.

8 Infrastructure, Compute and Data Foundations

- **MeluXina-AI & AI Factory.** A €60 million upgrade will add an AI-optimised supercomputer by 2H 2026 and create a "one-stop" AI Factory for SMEs, with 50 % of capacity reserved for EuroHPC collaboration.
- **Sovereign data & cloud.** Projects like Clarence provide air-gapped sovereign cloud services to regulators and ministries, reinforcing data autonomy.
- **Open-data roadmap & LU-CIX backbone.** Expanded datasets and six interconnected data-centres feed trusted data lakes for research and innovation.

With Bill 8476 nearing parliamentary adoption, a CNPD-centred oversight constellation, mandatory regulatory sandboxes and heavy investment in MeluXina-AI form a holistic, EU-compliant governance stack. As a result of weaving together ethical safeguards, sovereign infrastructure and real-world experimentation, Luxembourg is charting a pragmatic path to trustworthy, innovation-driven AI that matches its scale yet projects influence well beyond its borders.



Luxembourg's Approach to Fostering AI Innovation

Luxembourg's comprehensive approach to AI governance, deployment and innovation is rooted in human-centric values, resilience-based planning and EU-coordinated action. The Digital Decade National Strategic Roadmap for Luxembourg (October 2023; April 2024 update) translates these principles into quantified 2030 trajectories for skills, infrastructure and business take-up, pairing each target with concrete policy levers ranging from sovereign-cloud services to AI sandboxes.

1. Unlocking Economic Value through Targeted AI Deployment

The roadmap puts AI at the centre of value creation in finance, health-tech, space, smart mobility and clean-tech—all priority sectors in Luxembourg's diversification policy. Businesses already outperform the EU average on AI use (13 % vs 8 % in 2021), and the government sets the EU-level goal that 75 % of firms adopt AI, cloud and big-data solutions by 2030. Measures include dedicated edge-cloud nodes for smart-city and space-industry workloads and tax credits covering 18 % of CAPEX/OPEX for SME digital projects.



2. Catalysing Innovation through Strategic Clustering and Ecosystems

Luxembourg leverages a tight innovation web—University of Luxembourg/SnT, LIST, LIH, Luxinnovation, LHoFT and the Digital Innovation Hub—to channel R&D grants (~€200 m in five years) and accelerate SME pilots inside “living-lab” sandboxes mandated for every competent authority. The start-up roadmap From Seed to Scale identifies >500 active start-ups across fintech, health-tech and space, reinforcing the country's “test-bed” brand.

3. Enabling Trusted and Responsible Data Use

A sovereign GovCloud run by the Government IT Centre, the Luxembourg National Data Service and forthcoming sectoral data spaces (health, energy, smart cities) underpin secure, interoperable data exchange. The roadmap pairs these assets with GDPR-enforced oversight by the CNPD and commits €0.5 m annually to grow expertise in trustworthy data intermediaries.

4. Scaling AI in the Public Sector

Luxembourg is on track to hit 100 % digital public-service availability for citizens and businesses by 2030, thanks to the Guichet.lu and MyGuichet one-stop platforms. Agency pilots include sovereign LLM chatbots and algorithm registers that log high-risk systems, delivering transparency and accountability in line with EU AI-Act requirements.



5. Advancing Compute and Research Infrastructure

The MeluXina EuroHPC supercomputer anchors national compute strategy; a quantum-computer simulator and small European quantum computer will be attached from 2024. Edge-node roll-outs (17 forecast by 2030) feed latency-sensitive AI use-cases, while participation in IPCEI-CIS secures next-gen cloud-edge continuity.

6. Empowering SMEs and Start-ups

Initiatives such as Fit 4 Digital vouchers, cloud-application co-creation for SMEs and an Open-Source Programme Office aim to lift SME digital intensity to 90 % by 2030. The roadmap explicitly targets rapid uptake of ready-made LLM services to accelerate AI entry for small firms.

7. Developing a Future-Ready Workforce

Digital-skills measures span school coding curricula, Luxembourg Tech School, the Elements of AI MOOC (1 % of workforce), and labour-market programmes like Skills4Job and ICT Start. These efforts seek to raise residents with at least basic digital skills from 64 % to the EU 80 % target while tackling gender gaps through mentoring schemes such as Women Cyber Force.

8. Aligning AI Innovation with Sustainability Goals

Section 4 of the roadmap links AI, cloud and data-space initiatives to smart-energy systems, circular-economy IT procurement and low-latency 5 G corridors (5 G DeLux, 5 G Melusina), projecting digital tech as an enabler of the green transition.

9. Building Public Trust and Ethical AI

The roadmap re-affirms Luxembourg's commitment to a human-centric digital society, embedding privacy-by-design, disinformation counter-measures (EDMO Belux) and child-safety campaigns (BEE SECURE). Regulatory sandboxes, CNPD oversight and ISO/IEC standard-setting via ILNAS operationalise ethical AI principles.

10. Shaping Global AI Governance and Standards

Luxembourg co-drives EuroHPC, participates in 5 G cross-border corridors, IPCEI-CIS and Genome EDIC, and feeds findings into EU AI-Act implementation. This multi-country posture aligns national rules with pan-European standards on connectivity, cloud-edge portability and quantum security, projecting Luxembourg as a high-impact “living laboratory” for trustworthy AI far beyond its borders.





Figure 1: Human-centric approach adopted as covered in Luxembourg's National AI Strategy

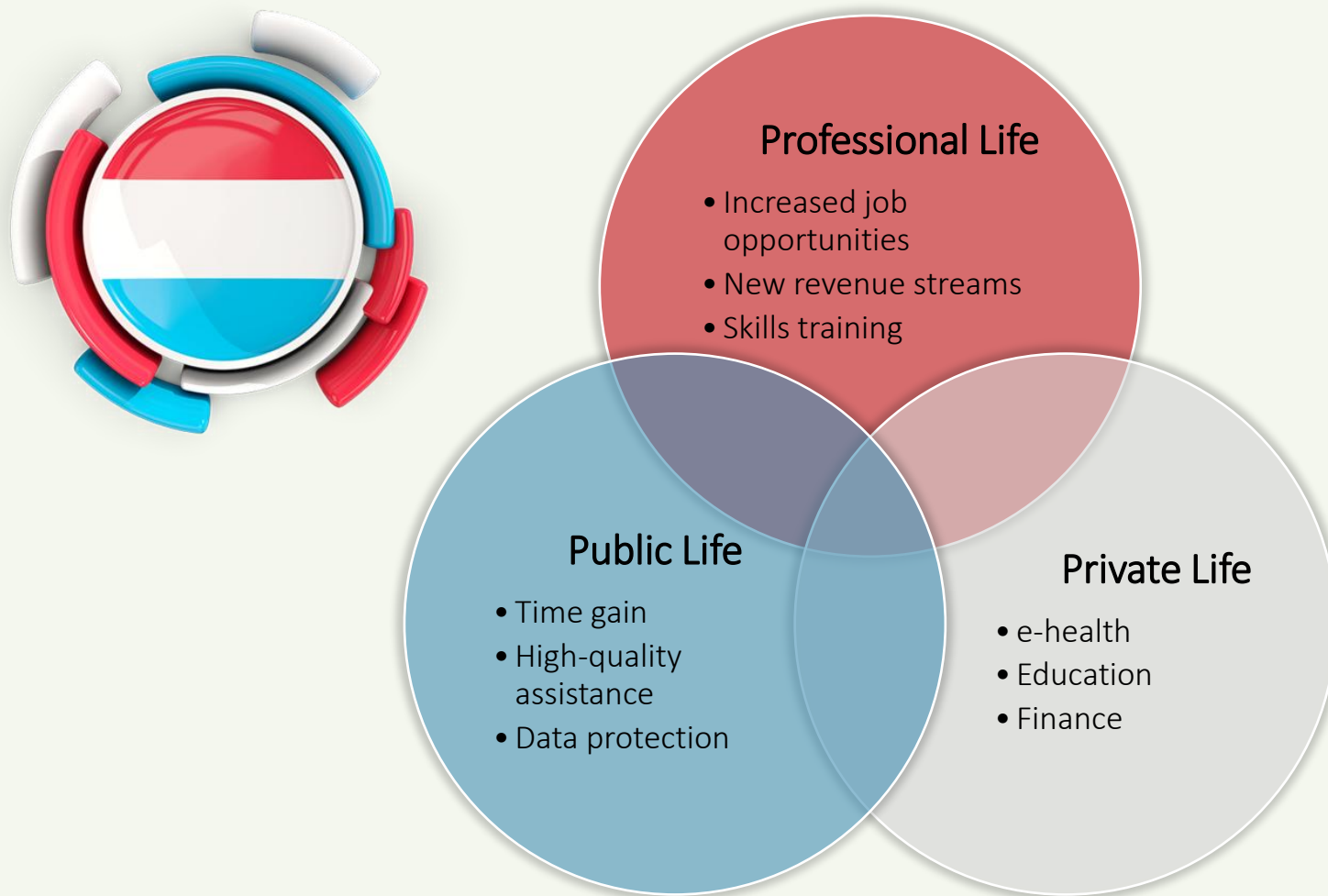
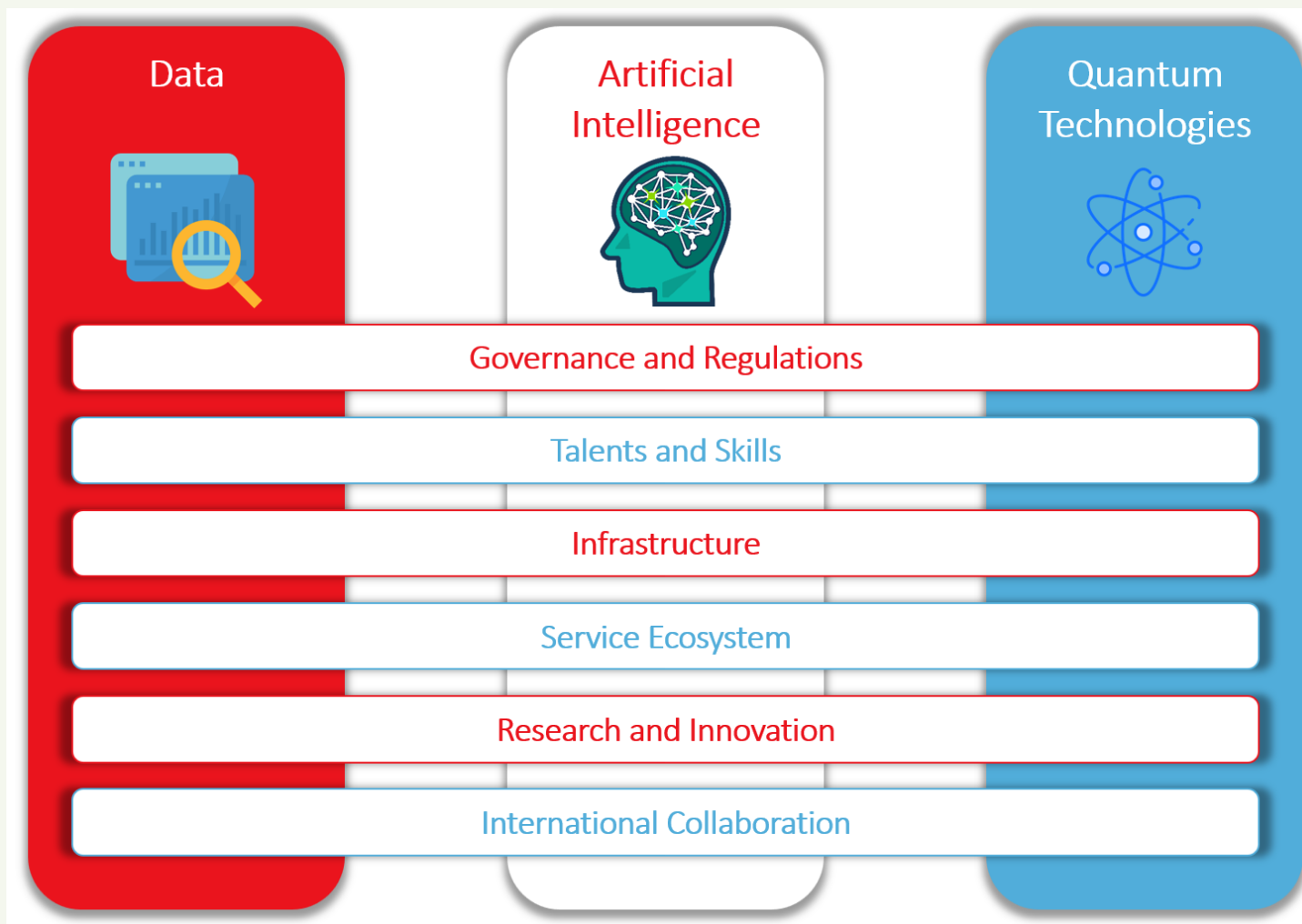




Figure 2: Luxembourg's vision in data, AI and quantum technologies





Digital Decade 2025: Country Report

Luxembourg continues to aim for a 75% adoption rate of Artificial Intelligence (AI) among businesses, aligning with the European Union's 2030 objective. To support this goal—particularly among small and medium-sized enterprises (SMEs)—the revised strategic plan introduces two new targeted initiatives

The first initiative, Fit 4 Digital – AI, launched in February 2025, is designed to assist SMEs in identifying potential AI applications, conducting feasibility assessments, and crafting tailored implementation strategies. The second, known as SME Packages – AI, is a new financial support program intended to co-finance tangible investments in AI technologies and services, in line with State aid regulations. These tools are strategically aligned to ensure the smooth translation of diagnostic insights into actionable plans. By building on pre-existing programmes, the new approach aims to overcome common adoption challenges such as cost barriers, limited expertise, and insufficient technical infrastructure. While the policy adjustments mark a significant shift and are consistent with the EU's 2030 benchmarks, their long-term effectiveness will rely heavily on continued outreach efforts, accessibility for smaller companies, and the emergence of successful AI implementation examples that foster trust in the technology. A notable advancement occurred in late 2024, when Luxembourg was chosen as the site of one of the EU's AI Factories. This selection is expected to propel the country's AI agenda forward, through the deployment of MeluXina-AI, a supercomputer specialized in AI tasks, and the establishment of a dedicated AI Factory focused on driving innovation in the coming years.

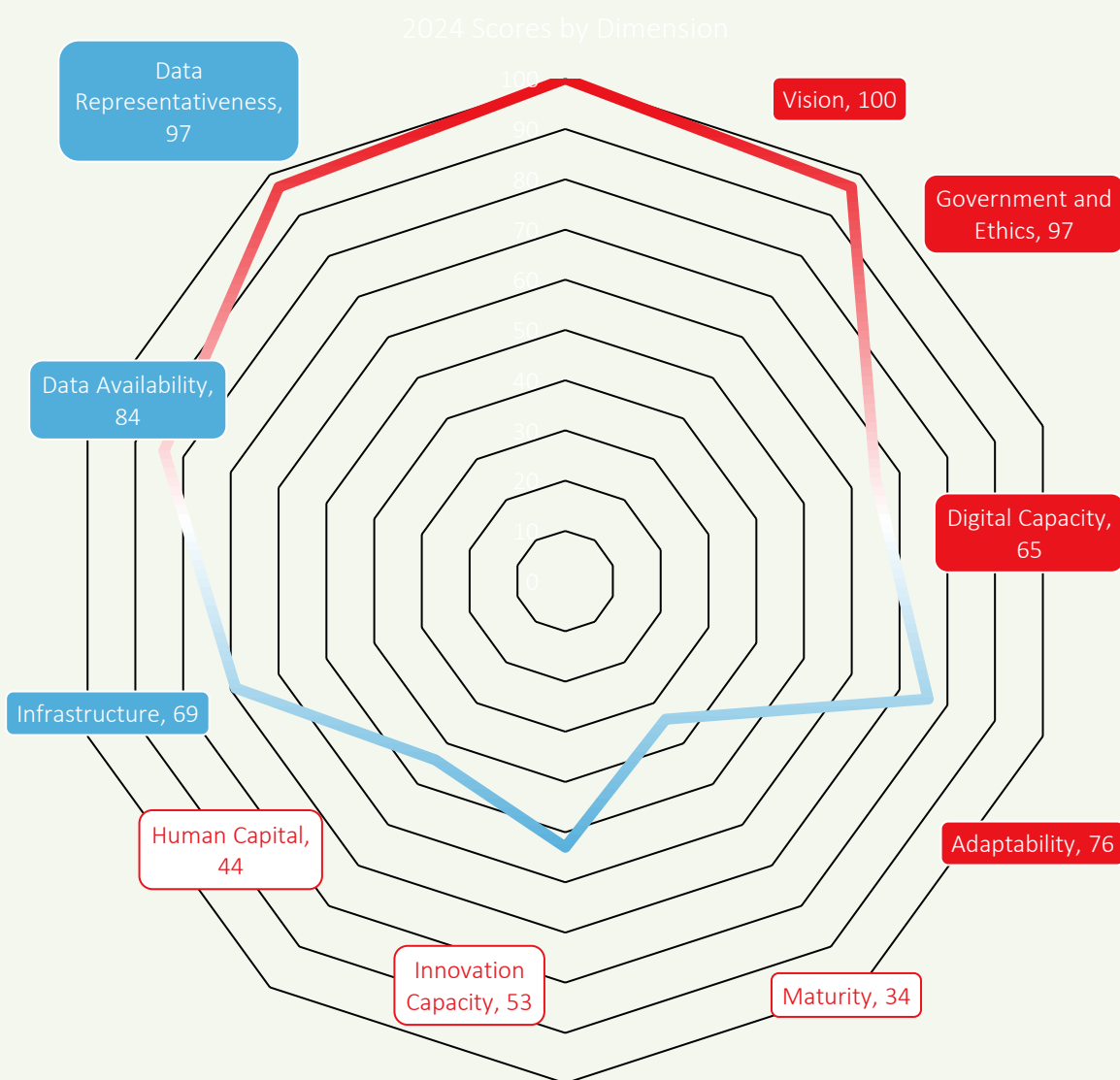
The 2025 Eurobarometer survey revealed that 85% of Luxembourg's population believes public institutions should lead the development of AI and other digital technologies, ensuring alignment with fundamental rights and values. This marks a significant increase—9 percentage points—over the previous year, highlighting a growing public interest in responsible AI development. In response to the 2024 recommendation concerning AI and data analytics, Luxembourg implemented meaningful policy measures during that year to enhance SME readiness for AI adoption, as detailed above. However, regarding data analytics specifically, no documented actions have yet been reported in relation to the recommendation.





Government AI Readiness by Oxford Insights

'Western Europe ranks as the highest-performing region in this year's Government AI Readiness Index, with five countries in the global top 10. France leads the regional ranking with a score of 79.36, narrowly ahead of the United Kingdom (78.88), followed closely by the Netherlands (77.23), Germany (76.90), and Finland (76.48). With an average regional score of 69.56, Western Europe outperforms the global average (47.59) across all three pillars of Government, Technology, and Data & Infrastructure. The region's standout strength is the Data & Infrastructure pillar, where it averages 81.91, more than 21 points above the global benchmark (59.62), reflecting its advanced digital infrastructure and high-quality data environment. Although a few microstates such as San Marino, Liechtenstein, and Andorra fall slightly below the global average in the Government pillar, Western Europe's dominance at the top of the index underscores its continued leadership in AI readiness.'





Government

Luxembourg records a Government-pillar score of 84.67, underscoring a highly proactive state apparatus that is already translating the EU AI Act into national practice. A forthcoming Law on the Implementation of the AI Act designates the data-protection regulator (CNPD) as the horizontal market-surveillance authority while mandating sectoral watchdogs—CSSF (finance), ILR (connectivity & media) and the Health Directorate—to run domain-specific AI sandboxes. Complementary strategies such as the Digital Decade National Strategic Roadmap and the 2019 AI Strategic Vision embed trustworthy-AI principles across public services and promote “living-lab” experimentation. The planned National Technology Ethics Committee will add an extra layer of rights-based oversight, ensuring that algorithmic transparency, non-discrimination and privacy remain central as AI systems move from pilot to production.

Technology Sector

A Technology-pillar score of 43.81 signals a dynamic—but still scaling—innovation ecosystem. Luxembourg’s tech strengths cluster around fintech, space, health-tech and cyber-security, boosted by institutes such as the University of Luxembourg’s SnT, LIST and the Luxembourg House of Financial Technology (LHoFT). Targeted schemes—FNR research grants, Fit 4 Start vouchers, and the Digital Innovation Hub’s SME coaching—aim to raise the share of firms developing or integrating AI solutions well above today’s 13 %. Continued expansion of sovereign cloud services and greater venture funding depth will be pivotal to convert R&D excellence into globally competitive AI scale-ups.



Data & infrastructure Sector

With a Data & Infrastructure score of 83.40, Luxembourg boasts one of Europe’s densest digital backbones. The MeluXina EuroHPC supercomputer delivers petascale compute for domestic SMEs and cross-border research, while six LU-CIX carrier-neutral data centres and near-universal 5 G/VHCN coverage ensure low-latency connectivity. The Luxembourg National Data Service federates public-sector datasets under GDPR-compliant access regimes, and forthcoming sectoral data spaces (health, finance, energy) provide the trusted environment needed for high-quality, interoperable AI training data.

A composite score of 70.63 places Luxembourg comfortably above the global average (47.59) and on par with Western Europe’s high performers. Outstanding public-sector readiness and first-rate data infrastructure form a solid platform; the next frontier is scaling the technology-innovation pillar by deepening the start-up pipeline, broadening sand-box participation and attracting growth-stage capital. If these efforts keep pace with the EU Digital Decade targets, Luxembourg is poised to consolidate its reputation as a compact yet influential “living laboratory” for trustworthy AI in Europe.



Conclusion

Artificial Intelligence (AI) is reshaping economies and governance worldwide, and Luxembourg is positioning itself as a nimble, ethics-first innovator in the European Union's AI ecosystem. Built on a foundation of digital sovereignty and EU-aligned legal infrastructure, Luxembourg's approach combines supervisory-led governance with rights-based safeguards and "living lab" experimentation. Rather than enacting standalone AI legislation, Luxembourg operationalizes the EU AI Act through Bill 8476, regulatory sandboxes, and strategic frameworks like its 2019 Strategic Vision and the 2025 Digital Sovereignty 2030 roadmap.**

At the heart of Luxembourg's AI governance model is the Commission nationale pour la protection des données (CNPd), which will serve as the national market surveillance authority and single point of contact under the EU AI Act. A network of sectoral authorities—such as the CSSF (finance), ILR (connectivity and media), ALMPS (health), and the CAA (insurance)—complements the CNPD and is mandated to run domain-specific AI sandboxes. This coordinated structure is supported by the CNPD's AI sandbox, "Sandkëscht," launched in May 2024, offering controlled testing environments where AI developers can collaborate with regulators and refine high-risk systems before deployment.

Luxembourg's institutional architecture is designed for agile, multi-actor collaboration. A forthcoming National AI Coordination Hub will harmonize oversight across domains, while public-sector initiatives such as the AI4Gov committee scout and scale AI use cases across ministries. Annual capacity-building plans and pooled technical labs ensure smaller authorities can effectively supervise complex AI models, reinforcing a whole-of-government commitment to trusted AI.

On the international stage, Luxembourg maintains strong alignment with EU-wide initiatives, including the Ethics Guidelines for Trustworthy AI, EuroHPC, and the Hiroshima AI Process. Cross-border testbeds such as the 5G DeLux and Melusina corridors—with France and Germany—demonstrate Luxembourg's commitment to globally interoperable standards. MeluXina, Luxembourg's EuroHPC supercomputer, underpins these efforts, offering AI-optimized compute power and reserving 50% of its capacity for European collaboration.

Aligned with the Digital Decade National Strategic Roadmap, Luxembourg targets key sectors such as finance, healthtech, mobility, and public services for AI transformation. Flagship instruments include sovereign cloud services like Clarence, the Luxembourg National Data Service (LNDS), and the Guichet.lu public service platform, which now pilots sovereign LLM chatbots and algorithm registers. These initiatives are supported by strategic funding tools like the Digital Tech Fund, Fit 4 Digital vouchers, and more than €200 million in AI-focused R&D investments.

Luxembourg's digital backbone reinforces its AI readiness. With near-universal 5G/VHCN coverage, six LUCIX data centers, and trusted sectoral data spaces, Luxembourg ensures GDPR-compliant, high-quality data environments for AI training and deployment. The MeluXinaAI upgrade and upcoming AI Factory will halve compute costs for SMEs and embed quantum capability into Luxembourg's supercomputing infrastructure.

Through this fusion of EU-compliant governance, sovereign data infrastructure, and dynamic experimentation, Luxembourg is building a resilient, future-ready model of AI oversight. By integrating human rights, regulatory agility, and scalable innovation, Luxembourg stands out as a compact yet influential actor in shaping the future of AI in Europe and beyond.





Appendix A: Expert Vignettes



Michael Godfriaux

Chiwodvice



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“Luxembourg is strategically positioning itself as Europe's premier 'living laboratory' for trustworthy AI in finance, leveraging its 'Sandkëscht' sandbox and sovereign data infrastructure to bridge the gap between the EU AI Act's principles and real-world implementation.”

”

Biography

Michael Godfriaux, Michael Godfriaux is a Brussels-based consultant specializing in AI strategy and regulatory compliance. With extensive experience in the public and financial sectors, he advises organizations on navigating the complexities of the EU AI Act and implementing robust AI governance frameworks.



AI & Partners

Amsterdam - London - Singapore



Geoffroy Hermanns

Norton Rose Fulbright



“

As AI reshapes the legal contours of the Luxembourg fund industry, it challenges the way the sector operates, but positively, by enhancing regulatory precision, streamlining compliance, and redefining the role of legal professionals.



”

Biography

Geoffroy Hermanns is an international asset management lawyer with over 15 years experience in funds, corporate and regulatory matters. He advises some of the world's leading financial institutions and asset management firms on the structuring and formation of a wide range of Shari'a-compliant and conventional investment funds, including private equity funds, real estate funds, infrastructure funds, hedge funds and retail/public funds, (regulated and unregulated), including hybrids. His practice covers the Middle East and Luxembourg with extensive experience in the DIFC (DFSA), ADGM (FSRA), Luxembourg (CSSF), federal UAE (SCA), Qatar (QFCRA), Bahrain, Saudi Arabia (CMA) and offshore (Cayman Islands, Jersey, BVI). Geoffroy also has extensive knowledge of the corporate and regulatory structuring and establishment of asset management and advisory businesses as well as carried interest arrangements



Appendix B: Strategic Commentary



Michael Godfriaux

Chiwodvice



1. The SME adoption challenge: beyond financials to talent and trust

Luxembourg's recent policy package—including **Fit 4 Digital – AI** and targeted tax credits—appropriately addresses cost barriers that hinder small and medium-sized enterprises (SMEs). Yet, from a practitioner's vantage point the decisive obstacles lie beyond balance-sheet constraints. Two interlocking frictions dominate SME decision-making: (i) an acute shortage of in-house data-science and ML engineering talent in an internationally competitive labour market; and (ii) the perceived legal and compliance complexity of operating under overlapping obligations (notably the EU AI Act, DORA and GDPR). For many small firms without dedicated legal or compliance resource, the regulatory landscape appears procedurally and reputationally risky — a “Triple Bind” that frequently outweighs the financial calculus for investing in AI.

Financial incentives therefore remain necessary but insufficient. The most effective interventions combine targeted subsidies with accessible, hands-on expertise that reduces regulatory uncertainty and builds operational capability. In practice, this means scaling the Digital Innovation Hub's coaching and technical assistance, embedding compliance navigation into voucher and grant programmes, and producing simplified, SME-focused compliance playbooks and templates. Such measures reduce the certainty and cost of compliance effort, lower the barrier to first pilots, and increase the likelihood that feasible, trustable AI use-cases move from experiment to production. Framed this way, Luxembourg's support ecosystem can materially accelerate SME adoption from the present 13% towards the Digital Decade objective of 75%.



2. “Sandkëscht” as a financial RegTech crucible

The **Sandkëscht** sandbox is already a distinctive pillar of Luxembourg’s living-lab approach. Its strategic value, however, extends beyond technical testing: it can operate as an arena for real-time regulatory de-confliction where financial innovators and supervisors resolve the tensions that arise at the intersection of the AI Act, DORA, GDPR and the AML Regulation. Financial AI use-cases often generate an explicit trade-off between transparency (required by the AI Act) and confidentiality (required by AML/financial secrecy rules)—a tension that, if unresolved, blocks safe deployment.

Sandkëscht is therefore uniquely suited to evolve from a testing facility into a **Joint Regulatory Guidance Protocol**. Under such an operating model, the CSSF, CNPD and FIU would coordinate synchronous, sandbox-specific guidance while sovereign-cloud platforms (e.g., the Clarence project) provide the trusted, auditable infrastructure necessary for sensitive processing. This coordinated forum would: (a) produce reconciled, implementable supervisory guidance for high-risk financial AI; (b) enable confidential, auditable experiments on air-gapped infrastructure; and (c) create precedent-based playbooks that smaller firms can reuse. Operationalised in this way, Sandkëscht would not only lower compliance friction for incumbent banks and fintechs, but also become a core competitive advantage for Luxembourg as a hub for RegTech and responsible financial innovation.

3. Operationalising MeluXina: posing the next-generation governance questions

The €60 million upgrade to **MeluXina** and the ambition to create a one-stop **AI Factory** for SMEs represents a materially strategic national investment. Announcing compute capacity is straightforward; governing it to maximise public value is the operationally harder task. To shift the debate from capability to durable public benefit, the report should surface the governance questions that will shape long-term outcomes:

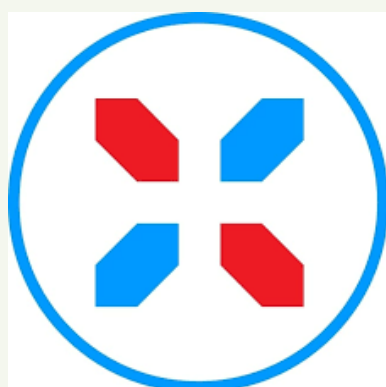


- **Access and equity.** What allocation and pricing model will prevent concentration of subsidised capacity by large incumbents while ensuring SMEs and early-stage innovators obtain meaningful access? What quotas, tiered pricing or application priorities will be used to balance domestic industrial policy with EuroHPC collaboration objectives?
- **Intellectual property.** Which IP regimes will apply to models, derivatives and datasets trained on MeluXina? How will ownership, licensing and commercialization rights be structured to incentivise private participation without undermining public-interest research?
- **Data sovereignty and cross-border collaboration.** How will the 50% of compute reserved for EuroHPC be governed to guarantee data sovereignty and lawful access controls, particularly where cross-border research and co-development raise jurisdictional questions? What contractual and technical safeguards (e.g., provenance, audit trails, enclave execution) will be mandated?
- **Operational transparency and accountability.** What reporting, audit and impact-assessment obligations will tenants face to ensure the facility delivers measurable public value and aligns with human-centric governance objectives?

Posing these questions does not represent criticism of the investment; rather, it elevates the analysis from descriptive to strategic. Early attention to allocation rules, IP frameworks and sovereignty safeguards will materially affect whether MeluXina becomes an inclusive enabler of SME innovation or an under-utilised asset concentrated in a few hands. By foregrounding governance design as a core component of infrastructure strategy, Luxembourg can maximise the public return on its AI investment and further consolidate its role as a compact yet influential “living laboratory” for trustworthy AI.



Appendix C: Critical Review of the EU AI Act and the Large AI Grand Challenge



Sourced from
Luxinnovation



This appendix critically examines the European Union's evolving policy and funding landscape for artificial intelligence (AI), focusing on the EU AI Act and associated initiatives such as the Large AI Grand Challenge, EuroHPC Joint Undertaking (JU), and AI Factories. The purpose is to contextualise these developments within Luxembourg's national AI ecosystem, evaluating their implications for governance, innovation capacity, and cross-border collaboration.

1. The EU AI Act: Governance and Risk Frameworks

The EU AI Act represents the world's first comprehensive legal framework for artificial intelligence, operationalising a risk-based regulatory model that distinguishes between minimal, limited, high, and unacceptable risk categories. Its enactment in 2024 signals a paradigmatic shift from voluntary ethical guidance toward enforceable compliance mechanisms.

While the Act provides much-needed regulatory certainty and strengthens public trust, its implementation poses significant challenges for small and medium-sized enterprises (SMEs) and startups—the core of Luxembourg's innovation ecosystem. Compliance burdens and conformity assessments may strain limited resources, potentially discouraging experimentation and agile innovation. Conversely, Luxembourg's regulatory expertise and digital governance infrastructure could be leveraged to position the country as a regional hub for AI compliance and certification, particularly through regulatory sandboxes and digital testing facilities.

2. The Large AI Grand Challenge: Scaling European AI Innovation

The Large AI Grand Challenge—supported by a €4 billion public-private investment—intends to accelerate the development of general-purpose AI (GPAI) models and expand access to high-performance computing (HPC) resources. This initiative embodies a strategic shift toward Europe's technological sovereignty by coupling ethical standards with competitive industrial development.



Critically, while the Challenge enhances access to computational infrastructure and funding, it risks reinforcing disparities between well-established AI ecosystems and smaller member states. Luxembourg, hosting EuroHPC JU, is uniquely positioned to benefit from proximity to these resources, yet must actively integrate its research institutions and startups into transnational project consortia to avoid peripheral participation. A national strategy that aligns AI R&D priorities with the Challenge's themes—such as multilingual AI and sustainable computing—could maximise domestic spillovers.

3. EuroHPC Joint Undertaking: Strategic Infrastructure for AI

EuroHPC JU, headquartered in Luxembourg, plays a central role in providing HPC and quantum computing capacity to European AI developers. Its €7 billion budget underpins Europe's ambition to achieve leadership in supercomputing and AI-driven innovation.

For Luxembourg, EuroHPC offers a tangible anchor to consolidate its role as a digital infrastructure node in Europe. However, the degree to which local enterprises and research organisations access and exploit this resource remains limited. Greater coordination between EuroHPC, the National Competence Centre in HPC, and innovation intermediaries such as Luxinnovation could improve knowledge transfer, capacity-building, and local utilisation of HPC assets for AI experimentation.

4. General-Purpose AI (GPAI): Ethical Standards and Innovation Tensions

The EU's emphasis on "trustworthy AI"—codified through ethical guidelines and the AI Act—aims to ensure transparency, human oversight, and accountability in GPAI systems. While conceptually robust, this normative approach may conflict with the speed and scale demanded by frontier AI research.

Luxembourg's participation in EU GPAI initiatives should therefore balance ethical alignment with pragmatic pathways for innovation. National research institutions and digital startups could serve as experimental sites for applying ethical-by-design principles in practical contexts, helping to operationalise EU-level standards through applied research and pilot projects.



5. The EU AI Office: Institutionalising Oversight and Support

The newly established EU AI Office, under DG CONNECT, will coordinate implementation and enforcement of the AI Act. Its mandate includes developing codes of practice, supporting regulatory sandboxes, and ensuring uniform compliance across member states.

This centralised governance model introduces consistency but may also create bureaucratic bottlenecks. For Luxembourg, effective engagement with the AI Office—potentially through seconded experts or liaison functions—could strengthen national influence over EU-level AI governance and provide early access to regulatory developments that affect domestic innovation activities.

6. AI Factories: Bridging Infrastructure and Talent

AI Factories are conceived as supercomputing service infrastructures that integrate AI-dedicated computational resources, data access, and human expertise. With €2.1 billion earmarked for AI-specific supercomputing upgrades, they represent a cornerstone of Europe's industrial AI capacity.

Constructively, while these Factories promise improved access to computational power for SMEs, their success depends on equitable participation and interoperability. Luxembourg's AI ecosystem could act as a testbed for inclusive access models, enabling smaller players to benefit from shared computational resources through coordinated national support schemes and open innovation partnerships.



7. European Digital Infrastructure Consortia (EDICs): Cross-Border Collaboration Mechanisms

EDICs formalise multi-country collaborations for digital infrastructure projects, including AI-relevant initiatives such as ALT-EDIC (language technologies) and CitiVERSE (urban digital twins). These mechanisms advance Europe's digital autonomy while promoting linguistic diversity and interoperability.

From a Luxembourgish perspective, EDICs offer a framework to scale domestic capabilities through participation in cross-border data and AI infrastructures. The country's multilingual and data-centric environment positions it well to contribute to projects like ALT-EDIC. However, sustained engagement and co-investment will be essential to ensure Luxembourg is not limited to observer status.

8. Critical Synthesis: Opportunities and Gaps for Luxembourg

Overall, the EU's AI policy architecture establishes an integrated model of regulation, funding, and infrastructure. For Luxembourg, these developments create both structural opportunities and strategic challenges:

- **Opportunities:** Leverage hosting of EuroHPC JU; position as a compliance and governance hub; embed Luxembourgish startups in EU innovation networks; exploit multilingualism for language AI.
- **Challenges:** Ensure equitable access to EU AI resources; prevent regulatory overreach from constraining smaller innovators; strengthen national coordination across ministries, research actors, and industry.



A coherent national strategy aligning Luxembourg's AI ecosystem with these European instruments could transform regulatory compliance into a competitive advantage. Proactive participation in initiatives such as AI Factories, the Large AI Grand Challenge, and EDIC consortia will be critical to ensure Luxembourg not only hosts but also shapes Europe's next generation of AI innovation.





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